

ROSHAN SHUKLA

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PROFESSIONAL SUMMARY

Software Engineer with experience building distributed systems, scalable backend services, and low-latency applications using Java, Spring Boot, Python, PostgreSQL, and Redis. Built production systems handling concurrent workloads, caching, fault tolerance, and event-driven architectures. Solved 500+ DSA problems with a peak LeetCode rating of 1784.

EDUCATION

B.Tech – Computer Science & Engineering (AI/ML)

Dr. A.P.J. Abdul Kalam Technical University (AKTU), Uttar Pradesh

Nov 2022 – Jun 2026

CGPA: 8.09 / 10

EXPERIENCE

Founding Software Engineer *SymptomWise*

Oct 2025 – Jan 2026

- Architected a multi-tenant healthcare platform supporting secure hospital-level data isolation and AI-powered medical triage workflows.
- Designed a dual-path execution architecture combining semantic caching and LLM inference, achieving a **61% cache hit rate** and reducing redundant inference workloads.
- Achieved **20ms p50 / 213ms p99** latency for cached requests and **680ms p50 / 1.53s p99** latency for inference workloads.
- Load tested the platform with **100 concurrent users**, sustaining **9.6 requests/sec** while processing **5800+ requests with zero failures**.
- Built deterministic validation and recovery pipelines ensuring reliability and consistent behavior under concurrent workloads.

Software Engineering Intern *GanakGyan Technologies*

Jun 2025 – Sep 2025

- Developed Java Spring Boot REST APIs supporting large-scale geospatial data processing and highly reliable retrieval workloads.
- Reduced API latency by **35% (300ms to 180ms)** through database indexing strategies, query optimization, and efficient connection pooling.
- Built stateless backend services and asynchronous ingestion pipelines processing **1000+ records per batch** with strict multi-threading constraints.
- Enhanced platform security and data protection reporting by implementing JWT authentication and role-based access control (RBAC).
- Utilized automated CI/CD pipelines and established logging, tracing, and metrics tooling to streamline production code deployments.

PROJECTS

IntervYou – Real-Time Collaborative Interview Platform

[Live](#) | [GitHub](#)

Next.js • Node.js • Redis Pub/Sub • WebRTC • Docker • Judge0

- Built a distributed real-time collaboration platform using WebSockets, Redis Pub/Sub, and PostgreSQL, achieving **110ms p50 / 337ms p95 latency** across **50 concurrent users**.
- Designed event-driven state synchronization with Redis-backed recovery mechanisms, preventing message loss during rigorous failure testing.
- Integrated secure containerized code execution using Docker and Judge0 API to ensure sandboxed, ad hoc code validation.

Automated Content Generation & Processing Pipeline

[GitHub](#)

Python • PyTorch • FFmpeg • SQLite • REST APIs • OAuth 2.0

- Engineered an automated, multi-stage Python data processing pipeline utilizing **PyTorch** transformers (Whisper) for sub-word alignment and context processing, reducing manual production times by **90%**.
- Implemented GPU-accelerated video rendering by wrapping **FFmpeg** sub-processes with hardware-accelerated encoding, managing dynamic subtitle compilation, audio scaling, and low-level kernel execution constraints.
- Developed secure API integrations via the YouTube Data API v3, incorporating a robust **OAuth 2.0** token lifecycle, automated token refresh validation, and an asynchronous queue scheduler to enforce strict execution concurrency safety.
- Built a SQLite-backed deduplication engine with retry handling and dynamic error-handling fallbacks to guarantee transactional data consistency.

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript, HTML, CSS

Backend & Frameworks: Spring Boot, REST APIs, Microservices, Django, Node.js

Databases & Storage: PostgreSQL, MySQL, MongoDB, Redis

Distributed Systems: Caching, Redis Pub/Sub, Event-Driven Architecture, Fault Tolerance, Idempotency, Stateless Services, Concurrency, Distributed Computing, Multi-Threading

Core CS: Data Structures, Algorithms, OOP, Object-Oriented Design, Operating Systems, Computer Networks, DBMS

Cloud, DevOps & Tools: Docker, AWS, Linux, Git, CI/CD, PyTorch

ACHIEVEMENTS

- Solved **500+ Data Structures and Algorithms problems** across LeetCode (Peak Rating: **1784**) and GeeksforGeeks.
- IEEE Student Branch Secretary, demonstrating resourcefulness and collaboration while managing a **15-member team** for technical initiatives.
- 2nd Runner-Up at **TechYuva 10.0** among competitive engineering cohorts.